

Mid Semester Examination 2025

Name of the Program: B.Tech

Semester: 3rd

Name of the Course: Electrical Machine-I

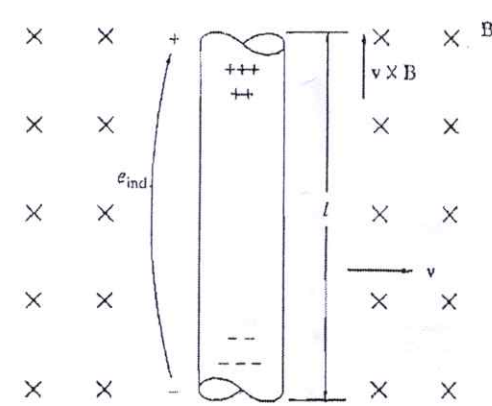
Course Code: **TEE302**

Time: 1-1/2 Hour

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Discuss hysteresis loop and explain hysteresis in terms of magnetic domain theory.	
OR		
(b)	What is magnetizing intensity? What is magnetic flux density? Discuss their relations?	
Q2	(10 marks)	CO2
(a)	Explain the difference between real and ideal transformer.	
OR		
(b)	Figure 1 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?  Fig.1	
Q3	(10 marks)	CO2
(a)	Fig 2 shows a conductor moving with a velocity of 10 m/s to the right in a magnetic field. The flux density is 0.5 T, out of the page, and the wire	

is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?

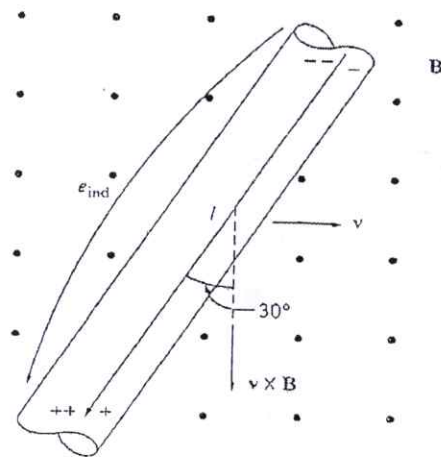


Fig.2

OR

(b) Discuss open circuit and short circuit test applied on single phase transformer with suitable diagram

Q4

(10 marks)

CO1

(a) Derive the relation for magnetic force and torque using co-energy

OR

(b) For the system shown below Fig.3 discuss energy balance equation.

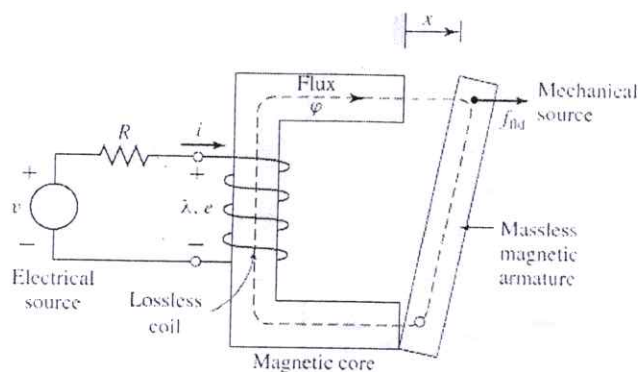


Fig.3

Q5

(10 marks)

(a) Discuss equivalent circuit of transformer

CO2

OR

- (b) A rectangular magnetic core with an air gap is shown in Fig.4 Calculate the exciting current needed to establish a flux density of $B_g = 1.2\text{T}$ in air gap given $N = 400$ turns and relative permeability of core is 4000

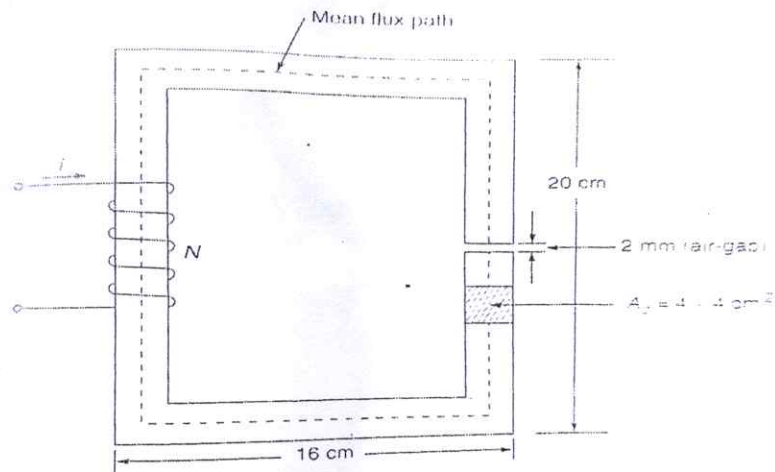


Fig.4

